FEAT-ZD-48600 · DE-05 Data Model Changes · Data Model change for the feature

Data Engineering agent · item feat.data_model_change · source hobasa-data-engineering-agent/outputs/FEAT-ZD-48600/feat.data_model_change.md · approved by Data engineer sim.data_engineer

feat.data_model_change: FEAT-ZD-48600

Data Model change for FEAT-ZD-48600 ("Show which overdue invoices belong to vendors that usually pay late"). This is step DE-05, Data Model Changes. The step has no human gate; automated checks, the independent judge and sampled human evaluation review it. The next step, DE-06 Data Mapping, reads this file.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	DE-05 Data Model Changes	record:brief
Owner	Data engineer	record:brief
Workbook action	Update the data model based on the source systems and ingested data	record:brief
Contracted inputs	task.components (7cb32bd), org.data_model (e8afce8)	record:brief
Background read	feat.design (722a98f): DC-1, DC-2, B1.2 to B1.4, DD-6, DD-7, NF-4	journey:feat.design
Other file read	`org/org.data_model.md` (organisation baseline, before the feature override)	file:hobasa-agentic-sdlc/org/org.data_model.md
Code read	`src/ap_pipeline/` (`ingest.py`, `transform.py`, `dq.py`, `pipeline.py`, `schedule.json`, `__init__.py`) from the step's tool view	tool:repo

1. Summary of the change

The **Invoice** entity gains one attribute, `paid_date (date or None)`, sourced from a new column `paid_date` that vendor feed v3 adds. Every other attribute stays as it is. Feed v2, which has no `paid_date` column, must keep loading (org.data_model; task.components section 2; journey:feat.design NF-4).

Two further, smaller changes follow from it. Neither is a new stored entity:

- The data-quality reject reasons gain one value, `'paid_date'` (task.components section 3; journey:feat.design B1.3).
- A derived, in-memory **Delay history** view is added: `{vendor: [delay_days, ...]}` (task.components section 4; journey:feat.design DC-2).

No storage, database table or schema migration is involved (journey:feat.design A1 "Data" row; task.components section 1 "Storage / Databricks: None").

2. Source system and ingested data (as read)

Item	Today	After this feature	Source
Source system	Vendor invoice feed: CSV text with a header row	Same source; feed v2 → v3 adds column `paid_date`. No new source, connection or feed	task.components (sections 1, 2)
Feed v2 header (sample)	`invoice_id,vendor,due_date,amount,paid`	Still accepted	task.components (section 1)

Item	Today	After this feature	Source
Feed v3 header	—	v2 columns plus `paid_date`	task.components (section 2); org.data_model
`paid_date` value in the feed	—	ISO date; empty when unpaid	record:memory (JOURNEY requirement: "optional paid_date column (ISO date, empty when unpaid)")
Ingestion (`ingest.parse_feed`)	`csv.DictReader`; one `dict` of strings per row, keyed by the header	No change: a v3 header carries `paid_date` into each row; a v2 header has no `paid_date` key	tool:repo (`ingest.py`); task.components (section 3)
Mapping to the model (`transform.to_model`)	Builds `id`, `vendor`, `due_date`, `amount`, `paid`	Adds `paid_date` (implemented in DE-08, TSK-01)	tool:repo (`transform.py`); task.components (section 4, row 1)

3. Data model: before and after

3.1 Invoice entity

Attribute	Type	Before	After	Change	Source
`id`	str	Yes	Yes	None	org.data_model
`vendor`	str	Yes	Yes	None (see open question DMQ-1 on "optional")	org.data_model; file:hobasa-agentic-sdlc/org/org.data_model.md
`due_date`	date	Yes	Yes	None	org.data_model
`amount`	Decimal, 2 dp	Yes	Yes	None. Decimal, never float	org.data_model; record:memory (D-2026-044, L-011)
`paid`	bool	Yes	Yes	None	org.data_model
`paid_date`	**date or None**	No	**Yes**	**Added** (feed v3)	org.data_model; journey:feat.design (DC-1)

Updated model line, as the feature override states it:

> Invoice: id (str), vendor (str), due_date (date), amount (Decimal, 2 dp), paid (bool), paid_date (date or None) [paid_date added in feed v3].

(org.data_model)

3.2 Attribute specification: `paid_date`

Property	Value	Source
Name in the model and in the feed	`paid_date`	org.data_model; task.components (section 2; CQ-DE4-4)
Type	Python `datetime.date`, or `None`	org.data_model; journey:feat.design (B1.3)
Nullable	Yes. `None` when the feed column is missing (feed v2), the cell is `None` (short row) or the cell is blank	journey:feat.design (B1.3, B1.4)
Format in the feed	ISO date, parsed with `date.fromisoformat` after stripping whitespace	journey:feat.design (B1.2); record:memory (JOURNEY requirement)
Invalid value	A non-blank value that is not an ISO date rejects the row with reason `'paid_date'`	journey:feat.design (B1.2, B1.4); task.components (section 4, row 1)

Property	Value	Source
Precedence	Validated only after the existing fields pass; an invalid existing field keeps today's reason (exception type name)	journey:feat.design (DD-6)
Relation to `paid`	Not cross-validated. A paid record without `paid_date` is kept; an unpaid row with a valid `paid_date` is accepted and keeps the date (interim, pending OQ-B6)	journey:feat.design (DD-7, B1.4)
Present on every record	Yes: every accepted record carries the `paid_date` key (a `date` or `None`)	task.components (section 4, row 1); journey:feat.design (B1.2 step 3)

3.3 Rejected row and data-quality report

Structure	Change	Source
Rejected row `{"row": dict, "reason": str}`	Shape unchanged; one new `reason` value, `"paid_date"`	journey:feat.design (B1.3); task.components (section 4, row 1)
`quality_report` → `reject_reasons`	No code change; `'paid_date'` appears in the sorted set when such a row is rejected	task.components (section 3, `dq.quality_report`); tool:repo (`dq.py`)

3.4 Derived view: Delay history (not stored)

Property	Value	Source
Shape	`Dict[str, List[int]]`, `{vendor: [delay_days, ...]}`	journey:feat.design (DC-2, B1.3)
Rule	`delay_days = (paid_date - due_date).days`, calendar days	task.components (section 4, row 2); record:memory (D-2026-031)
Included records	Paid records with a `paid_date` only	task.components (section 4, row 2); journey:feat.design (DC-2)
Order	Feed order within a vendor; vendors in first-seen order	task.components (section 4, row 2)
Negative delays (early payment)	Kept	task.components (section 4, row 2); journey:feat.design (DC-2)
Produced by	`delays.delays_by_vendor(records)` (new module, TSK-02)	task.components (section 4, row 2)
Consumer	ap_forecast `vendor_risk(history)`, one vendor's list at a time	journey:feat.design (DC-2)

3.5 Knowledge-graph view

```
flowchart LR
  F["Vendor invoice feed v2 / v3 (CSV)"] -- "column paid_date (v3 only)" --> I["Invoice<br/>id, vendor, due_date, amount
  F -- "invalid paid_date" --> R["Rejected row<br/>reason 'paid_date' (NEW value)"]
  R --> Q["Data-quality report<br/>reject_reasons"]
  I -- "paid and paid_date set" --> D["Delay history (derived, NEW)<br/>{vendor: [delay_days]}"]
  D --> V["ap_forecast vendor_risk"]
  I -- "DC-1 records" --> A["ap_aging overdue_with_risk<br/>(ignores paid_date)"]
```

Sources: task.components (section 4 diagram); journey:feat.design (A2, DC-1, DC-2, B3.3).

4. Compatibility

Consumer / case	Effect	Source
Feed v2 (no `paid_date` column)	Loads as today; every record has `paid_date = None`	journey:feat.design (NF-4, B1.4); task.components (section 2)
Existing record keys	Unchanged; `paid_date` is additive	journey:feat.design (DC-1)

Consumer / case	Effect	Source
`to_model` signature and `(records, rejected)` return shape	Unchanged	task.components (section 4, row 1)
ap_aging (DC-1 consumer)	Reads `id`, `vendor`, `due_date`, `amount`, `paid`; extra key `paid_date` ignored	journey:feat.design (DC-1, B3.3)
Schedule	Unchanged: daily, 02:00, Asia/Kolkata	tool:repo (`schedule.json`); task.components (section 3)

5. Code in ap_pipeline

This step changes no code. The model change is implemented by TSK-01 in DE-08 (`transform.to_model` with `_parse_paid_date` and `PAID_DATE_REASON`) and TSK-02 (`delays.delays_by_vendor`), as task.components section 4 plans. At that point the `transform.py` module docstring, which lists the model as "(id, vendor, due_date, amount, paid)", should also name `paid_date` (tool:repo `transform.py`; task.components section 4).

Current behaviour that the new attribute builds on (tool:repo `transform.py`):

- A missing `invoice_id`, `vendor`, `due_date` or `amount` key raises `KeyError`, and the row is rejected.
- `paid` is read with `r.get("paid", "false")`, so a missing `paid` column means unpaid.
- `amount` is `Decimal(r["amount"])`, with no check on the number of decimals.

`pipeline.run` calls `to_model` on the output of `parse_feed` and passes its result to `quality_report` (tool:repo `pipeline.py`). The repository is on `feature/story` at `3fcbe1b` (task.components).

6. Handover to DE-06 (Data Mapping)

Source column (feed)	Model attribute	Rule
`invoice_id`	`id`	Unchanged (strip)
`vendor`	`vendor`	Unchanged (strip)
`due_date`	`due_date`	Unchanged (ISO date)
`amount`	`amount`	Unchanged (Decimal)
`paid`	`paid`	Unchanged (`"true"`, case-insensitive, after strip; missing → false)
`paid_date` (v3, optional)	`paid_date`	**New**: missing, `None` or blank → `None`; ISO date → `date`; otherwise reject with reason `'paid_date'`, after the existing checks pass

Sources: tool:repo (`transform.py`); journey:feat.design (B1.2, B1.4); org.data_model.

7. Open questions (non-blocking)

ID	Question	Interim position	Owner (proposed)
DMQ-1	The organisation baseline says `vendor (str, optional)`, but the feature override says `vendor (str)`. Today's code rejects a row with no `vendor` column (`KeyError`), and ap_aging raises `KeyError` for a missing `vendor` (DD-8). Which definition is authoritative after this feature?	Follow the feature override (the contracted input). This feature does not change `vendor`	Data modeler

ID	Question	Interim position	Owner (proposed)
DMQ-2	Who updates `org/org.data_model.md` with `paid_date` once the feature is released, and when? The override sits only under `features/FEAT-ZD-48600/org-overrides/`	Treat the feature override as the model for this feature	Data engineering lead
DMQ-3	The model says `amount (Decimal, 2 dp)`, but `to_model` does not enforce 2 dp. This gap exists today and this feature does not add to it; ap_aging quantises downstream (DD-10)	No change in this feature	Data modeler
CQ-DE4-1 (carried)	When does feed v3 reach production? ZD-48470 brings feed v2	Build and test on synthetic v3 rows; v2 keeps loading	Product manager / Data engineering lead
CQ-DE4-4 (carried)	No feed v3 specification is in the inputs; the column name `paid_date` comes from the data model and the design	Use `paid_date`	Data modeler
OQ-B6 (carried)	Should an unpaid row with a valid `paid_date` be accepted?	Accepted, date kept (DD-7)	Product manager

No blocking questions.